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MATHEMATICS 2270
Midterm 1, Fall semester 2012

Name:.....**KEY**.....

1. For which b are $(1, b, 1)$, $(1, 1, -1)$, $(2, b+1, b-2)$ a basis of $\mathbb{R}^3$?

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & b+1 & b-2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & b-2 \end{bmatrix} \Rightarrow 3 \text{ pivots iff } b \neq 1, 2$$

2. Find all solutions to $x_1 + 2x_2 + 3x_3 + x_5 = 1$, $x_1 + 2x_2 + 3x_3 + x_4 + x_5 = 2$, $2x_1 + 4x_2 + 6x_3 - 2x_4 + 2x_5 = 0$.

$$\begin{array}{ccc|c} 1 & 2 & 3 & 0 & 1 & 1 \\ 1 & 2 & 3 & 1 & 1 & 2 \\ 2 & 4 & 6 & -2 & 2 & 0 \end{array} \rightarrow \begin{array}{ccc|c} 1 & 2 & 3 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -2 & 0 & -2 \end{array} \rightarrow \begin{array}{ccc|c} \textcircled{1} & 2 & 3 & 0 & 1 & 1 \\ 0 & 0 & 0 & \textcircled{1} & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + v \begin{pmatrix} -3 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + w \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

special soln matrix

$$\begin{bmatrix} -2 & -3 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Let $V = C(\mathbb{R})$ be the space of all continuous functions on $\mathbb{R}$ and $P = \mathbb{R}[x]$ be the space of all polynomials. Show that P is a subspace of V .

(1) 0 is a polynomial ✓

$$(2) (a_0 + a_1x + \dots + a_nx^n) + (b_0 + b_1x + \dots + b_mx^m)$$

$$= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + (a_3 + b_3)x^3 + \dots$$

The sum of 2 polynomials is a polynomial. ✓

$$(3) \lambda (a_0 + a_1x + \dots + a_nx^n) = (\lambda a_0) + (\lambda a_1)x + \dots + (\lambda a_n)x^n$$

Any scalar multiple of a polynomial is a polynomial.

4. Let $P = \mathbb{R}[x]$ be the space of all polynomials. Show that the set of polynomials $p(x)$ such that $d^3(p(x))/dx^3 = 0$ is a subspace of $\mathbb{R}[x]$, but the set of polynomials $p(x)$ such that $d^3(p(x))/dx^3 = 1$ is not a subspace of $\mathbb{R}[x]$.

$$d^3(p(x))/dx^3 = 1 \quad \text{Not a subspace as } \frac{d^3(0)}{dx^3} = 0 \neq 1$$

So 0 is not in this subspace.

$$d^3(p(x))/dx^3 = 0 \quad (1) \quad \frac{d^3(0)}{dx^3} = 0 \Rightarrow \text{contains } 0 \quad \checkmark$$

$$(2) \frac{d^3(p(x))}{dx^3} = 0 \quad \frac{d^3(q(x))}{dx^3} = 0 \Rightarrow \frac{d^3(p(x) + q(x))}{dx^3} = \frac{d^3p(x)}{dx^3} + \frac{d^3q(x)}{dx^3}$$

$$= 0 + 0 = 0.$$

$\Rightarrow$ closed under addition ✓

$$(3) \frac{d^3(p(x))}{dx^3} = 0 \Rightarrow \frac{d^3(\lambda p(x))}{dx^3} = \lambda \frac{d^3(p(x))}{dx^3} = \lambda \cdot 0 = 0$$

$\Rightarrow$ closed under scalar multiplication.

5. Find a system of equations with solution set $(1, -1, 1, -1) + t(2, 1, 0, 3)$.

$$\left[\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 3 \\ 0 & 3 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right]$$

$$\begin{cases} x_1 - 2x_2 = 3 \\ 3x_2 - x_4 = -2 \\ x_3 = 1 \end{cases}$$

6. Find the $A = LU$ decomposition for

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

(L means Lower triangular)
" " " " upper " "

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 1 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow{L_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow{L_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 3 \end{bmatrix} \xrightarrow{U}$$

$L_1 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$

$$L_2 L_1 A = U$$

$$A = L_1^{-1} L_2^{-1} U$$

$$L = L_1^{-1} \cdot L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$